Year 9 Mathematics

Unit 1: Number

1.5 STEM Calculating with Standard Form

STEM Calculating with Standard Form

STARTER · 5 MIN

Date: _____ **Title: STEM Calculating with Standard Form**

LEARNING OBJECTIVES

- Multiply and divide numbers in standard form
- Add and subtract numbers in standard form
- Use a calculator to solve problems in standard form

STARTER *Answer in your book. No calculators.*

1. Write 59 000 in standard form
2. Write 0.0601 in standard form
3. Simplify $10^{-3} \times 10^{2}$
4. Simplify $10^{3} \div 10^{-1}$

Starter — Answers

STARTER

Tick or correct in green pen

1. 5.9×10^{4}
2. 6.01×10^{-2}
3. 10^{-1}
4. 10^{4}

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can multiply and divide numbers in standard form	☐
SILVER	I can add and subtract numbers in standard form	☐
GOLD	I can use a calculator to solve problems in standard form	☐

Tick each box in the plenary when you can do it on your own.

Introduction — STEM Calculating with Standard Form

INTRODUCTION · 3 MIN

Deal with the numbers and the powers of 10 separately.

Multiply

Multiply numbers, add powers.

$(3 \times 10^4) \times (2 \times 10^3) = 6 \times 10^7$

Divide

Divide numbers, subtract powers.

$(8 \times 10^6) \div (2 \times 10^2) = 4 \times 10^4$

Add or subtract

Make the powers the same first.

$3 \times 10^4 + 5 \times 10^3 = 3.5 \times 10^4$

Check A

If $A \geq 10$, change it.

$10.8 \times 10^5 = 1.08 \times 10^6$

Examples 1 and 2 — Multiplying and Dividing

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint
×: multiply A, add the powers
÷: divide A, subtract the powers

Example 1

Work out

$(2.7 \times 10^{3}) \times (4 \times 10^{-5})$

in standard form.

Example 2

Work out

$$\frac{1.2 \times 10^{6}}{3 \times 10^{2}}$$

in standard form.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Multiply numbers, add powers

$2.7 \times 4 = 10.8, \quad 3 + (-5) = -2$

10.8×10^{-2}

$= 1.08 \times 10^{-1}$

Tip: 10.8 is not between 1 and 10.

Example 2

Divide numbers, subtract powers

$1.2 \div 3 = 0.4, \quad 6 - 2 = 4$

0.4×10^{4}

$= 4 \times 10^{3}$

Tip: 0.4 is less than 1, so A = 4.

Task 1

PRACTICE · 9 MIN

BRONZE *Give your answers in standard form*

Hint
×: multiply A, add the powers
÷: divide A, subtract the powers

1 $(1.2 \times 10^2) \times (3 \times 10^3)$

2 $(1.5 \times 10^5) \times (5 \times 10^3)$

3 $(4 \times 10^4) \times (6.25 \times 10)$

4 $\dfrac{8 \times 10^5}{2 \times 10^3}$

5 $\dfrac{1.2 \times 10^6}{3 \times 10}$

★ $(2 \times 10^5) \times (6 \times 10^{-2})$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 3.6×10^5

2 7.5×10^8

3 2.5×10^6

4 4×10^2

5 4×10^4

★ 1.2×10^4

Examples 3 and 4 — Adding and Subtracting

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint write as ordinary numbers first then change back to standard form

Example 3

Work out

$4.5 \times 10^4 + 3.2 \times 10^3$

in standard form.

Example 4

Work out

$6 \times 10^{-3} - 4 \times 10^{-4}$

in standard form.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Write as ordinary numbers

45 000 + 3200

= 48 200

= 4.82×10^{4}

Tip: never add the powers.

Example 4

Write as ordinary numbers

0.006 − 0.0004

= 0.0056

= 5.6×10^{-3}

Tip: line up the decimal points.

Task 2

PRACTICE · 9 MIN

SILVER

Give your answers in standard form

Hint write as ordinary numbers first then change back to standard form

1 $3 \times 10^4 + 2 \times 10^4$

2 $5 \times 10^3 + 4 \times 10^2$

3 $8.1 \times 10^5 - 6 \times 10^4$

4 $2.6 \times 10^6 + 9 \times 10^5$

5 $5 \times 10^{-2} - 3 \times 10^{-3}$

★ $7.2 \times 10^{-4} + 8 \times 10^{-5}$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 5×10^{4}

2 5.4×10^{3}

3 7.5×10^{5}

4 3.5×10^{6}

5 4.7×10^{-2}

★ 8×10^{-4}

Examples 5 and 6 — STEM Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: use a calculator

Hint brackets round each number
1 cm = 10^{-2} m 1 mm = 10^{-3} m

Example 5

(2 marks)

Work out

$(9.6 \times 10^{7}) \times (6.41 \times 10^{3})$

Give your answer in standard form to 3 s.f.

Example 6

(3 marks)

A hair is 8×10^{-5} m wide.

How many hairs side by side make 1 cm?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Calculator: use the ×10ˣ key

$9.6 \times 6.41 = 61.536$

61.536×10^{10}

$= 6.15 \times 10^{11}$ (3 s.f.)

Tip: put brackets round each number.

Example 6

1 cm = 0.01 m = 1×10^{-2} m

$10^{-2} \div (8 \times 10^{-5})$

= 125 hairs

Tip: change to the same units first.

Task 3

PRACTICE · 9 MIN

GOLD

Use a calculator. Standard form to 3 s.f. where needed.

Hint brackets round each number
1 cm = 10^{-2} m 1 mm = 10^{-3} m

1 $(4.7 \times 10^{6}) \times (3.2 \times 10^{5})$

2 $(3.5 \times 10^{2}) \div (6 \times 10^{9})$

3 $(8.3 \times 10^{-4}) \times (7.9 \times 10^{2})$

4 $\dfrac{7.2 \times 10^{5}}{2.7 \times 10^{-3}}$

5 A grain is 6×10^{-4} m wide.
How many fit across 3 cm?

★ A cell is 7×10^{-6} m wide.
How many fit across 1 mm?

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. 1.50×10^{12}
2. 5.83×10^{-8}
3. 6.56×10^{-1}
4. 2.67×10^{8}
5. 50

★ 143 (3 s.f.)

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Work out $(4 \times 10^3) \times (2 \times 10^5)$

SILVER Work out $7 \times 10^4 + 5 \times 10^3$

GOLD A cell is 2×10^{-5} m wide. How many fit across 1 mm?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE 8×10^8

SILVER 7.5×10^4

GOLD 50

Exam Practice — Multiplying and Dividing

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint
×: multiply A, add the powers
÷: divide A, subtract the powers

Question 1

(2 marks)

Work out $(5 \times 10^{4}) \times (3 \times 10^{-7})$

Give your answer in standard form.

.....

Question 2

(2 marks)

Work out

$$\frac{9 \times 10^{9}}{4 \times 10^{4}}$$

Give your answer in standard form.

.....

Exam Practice — Adding and Subtracting

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint write as ordinary numbers first then change back to standard form

Question 3

(2 marks)

Work out $6.4 \times 10^5 + 3 \times 10^4$

Give your answer in standard form.

.....

Question 4

(2 marks)

Earth is 1.5×10^8 km from the Sun.

Mars is 2.28×10^8 km from the Sun.

Find the difference in standard form.

..... km

Exam Practice — STEM Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint brackets round each number
1 cm = 10^{-2} m 1 mm = 10^{-3} m

Question 5

(2 marks)

Earth's mass is 5.97×10^{24} kg.
The Moon's mass is 7.35×10^{22} kg.
How many times heavier is the Earth?
Give your answer to 3 s.f.

.....

Question 6

(3 marks)

A red blood cell is 7.5×10^{-6} m wide.
How many cells side by side
make a line 3 cm long?

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

15×10^{-3} — M1

$= 1.5 \times 10^{-2}$ — A1

Tip: 4 + (−7) = −3

Question 2

$9 \div 4$ and $10^{9} \div 10^{4}$ — M1

$= 2.25 \times 10^{5}$ — A1

Tip: $9 \div 4 = 2.25$, $10^{9} \div 10^{4} = 10^{5}$

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

640 000 + 30 000 — M1

= 6.7×10^5 — A1

Tip: never add the powers.

Question 4

$2.28 \times 10^8 - 1.5 \times 10^8$ — M1

= 7.8×10^7 km — A1

Tip: 0.78×10^8 is not standard form.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$5.97 \times 10^{24} \div (7.35 \times 10^{22})$ — M1

= 81.2 (3 s.f.) — A1

Tip: use brackets around the bottom.

Question 6

$3 \text{ cm} = 3 \times 10^{-2} \text{ m}$ — B1

$3 \times 10^{-2} \div (7.5 \times 10^{-6})$ — M1

= 4000 cells — A1

Tip: change cm to m first.